Course Information

Essential details about the course

Key Information

Basic course details

Course Code

The course code is COMP3006E, facilitating easy identification.

Course Name

It's called "机器学习" in Chinese and "Machine Learning" in English.

Total Credit Hours

The course offers a total of 48 credit hours, with 32 theoretical and 16 experimental.

Offering Department

The School of Computer Science and Technology is responsible for offering this course.

Target Students

This course is designed for international undergraduate students.

Offering Semester

It is offered in the 3rd Semester during the autumn term.

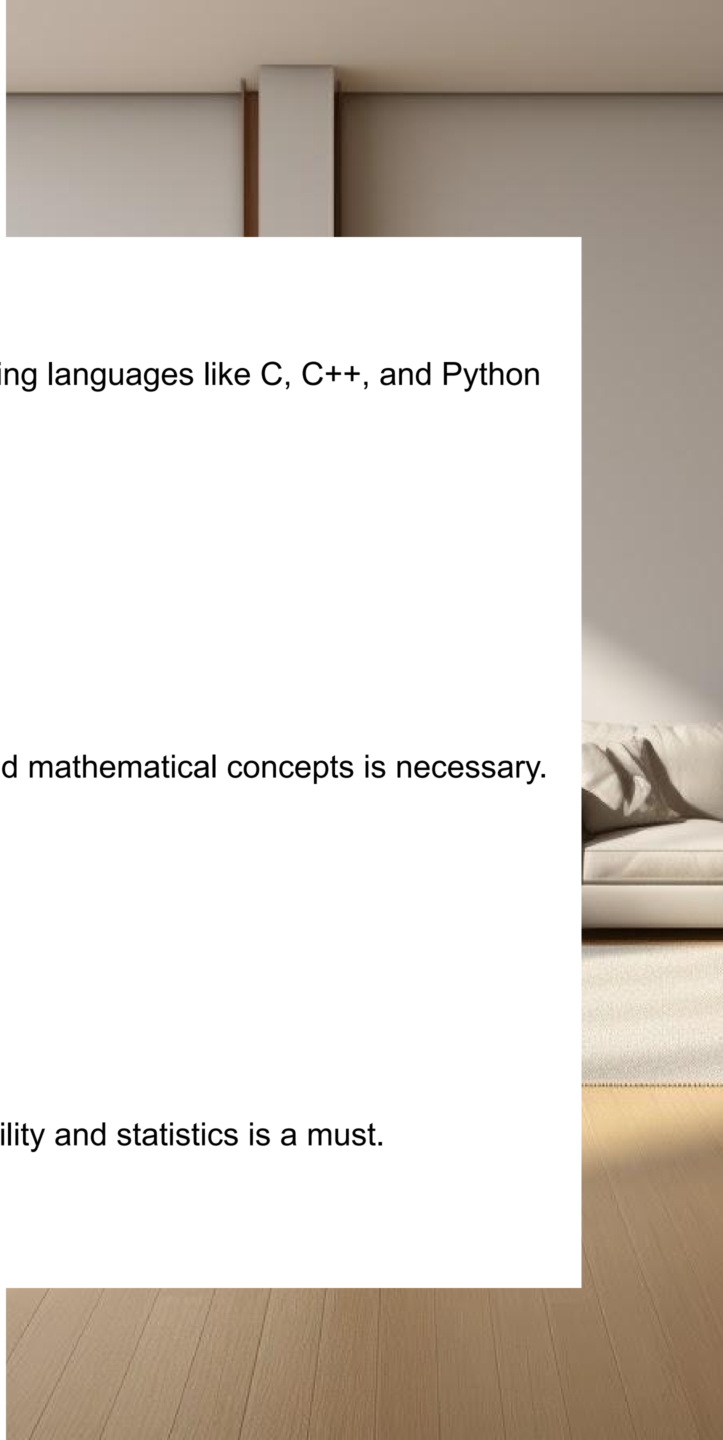
Time and Contact

The course starts on 2025 - 08 - 31, and the QQ Group ID is available for communication.

Required Knowledge

Necessary prior knowledge

- **Advanced Programming Languages**: Knowledge of programming languages like C, C++, and Python is required.
- **Advanced Mathematics**: A good grasp of advanced mathematical concepts is necessary.
- **Probability Theory and Mathematical Statistics**: Understanding of probability and statistics is a must.



Objective 1

Master the basics

1

Basic Concepts and Principles

Students will understand and master the basic concepts and principles of machine learning.

2

Theoretical Foundations

They will learn about the theoretical foundations of the machine learning field.

3

Research Hotspots

Get to know the current research hotspots in machine learning.

4

Development Trends

Be aware of the development trends in this field.

Objective 2

Grasp classical algorithms

Classical Machine Learning Algorithms

Students will understand and master classical machine learning algorithms.

Algorithm Characteristics

Comprehend the characteristics of each algorithm.

Suitable Engineering Problems

Know which engineering problems each algorithm is suitable for solving.

Transforming Problems

Develop the ability to transform practical problems into machine learning problems.

Objective 3

Comprehensive Analysis

Cultivate the ability to comprehensively analyze practical problems.

Cultivate the ability to comprehensively analyze practical problems.

Be able to design solutions for practical problems.

Be able to design solutions for practical problems.

Conduct experimental analysis and verification.

Conduct experimental analysis and verification.

Acquire the ability to solve practical engineering problems.

Acquire the ability to solve practical engineering problems.

Solve practical problems

Solve practical problems

Classroom Teaching

Teaching methods in the classroom

Heuristic Teaching	Auxiliary Tools	Case - based Teaching	Interactive Teaching	Experimental Projects
A heuristic teaching method is adopted to stimulate students' interest in learning.	Multiple auxiliary tools like teaching models and code - based courseware are used.	Case - based teaching is used to integrate theory with engineering practice.	Interactive teaching is implemented to increase face - to - face communication.	Interesting experimental projects are designed to attract students' attention.

Regression & Classification Models

Foundational predictive modeling techniques

1	**Linear and Logistic Regression**	Explore foundational regression techniques.
2	**Model Evaluation Metrics**	Assess performance using key metrics.
3	**Classification Techniques Overview**	Survey common classification methods.
4	**Handling Imbalanced Datasets**	Strategies for working with uneven data.
5	**Decision Trees and Bayesian Learning**	Introduce probabilistic modeling approaches.
6	**Naive Bayes Classifier Application**	Use in simple yet effective classification tasks.
7	**Tree - based Model Tuning**	Optimize for improved predictive accuracy.

Advanced Neural Networks

Exploring deep learning and its practical applications

Neural Network Fundamentals

Understand the basic architecture and functions of neural networks.

Convolutional Neural Networks (CNNs)

Explore how CNNs revolutionize image recognition tasks.

Recurrent Neural Networks (RNNs)

Learn about RNNs and their applications in sequence data processing.

Transformers in NLP

Discover how transformers enhance natural language processing tasks.

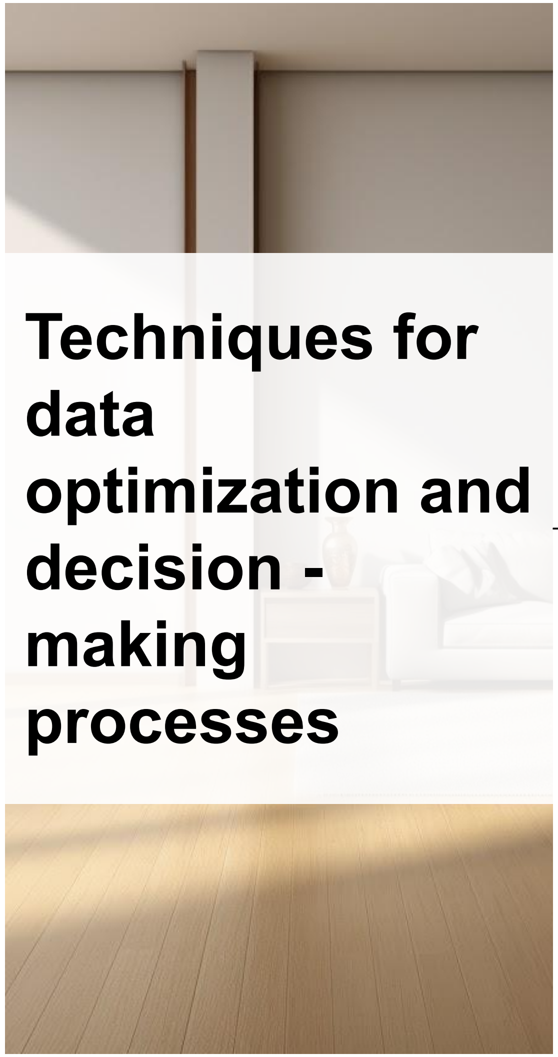
Practical Applications of Neural Networks

Real - world examples of deep learning in various industries.

Challenges in Neural Network Implementation

Common obstacles faced when deploying deep learning solutions.

Dimensionality Reduction & Reinforcement Learning



Techniques for data optimization and decision - making processes

1	**Principal Component Analysis (PCA)**	Learn to reduce dataset dimensionality effectively.
2	**Reinforcement Learning Basics**	Understand core concepts of reward - driven learning.
3	**Q - Learning and Deep Q - Networks**	Explore model - free learning algorithms.
4	**Real - World Implications**	Discover applications in autonomous systems.
5	**Advanced Dimensionality Reduction Techniques**	Investigate beyond PCA, like t - SNE.
6	**Implementing Reinforcement Learning Models**	Practical steps for developing RL solutions.
7	**Ethical Considerations in RL**	Address ethical issues in autonomous decision - making.

Experimental Projects

Details of experimental projects

1. Experimental Projects

The course sets 4 experimental projects totaling 16 credit hours.

2. Team Work

Students are required to complete experiments in independent teams.

3. Experimental Reports

They need to write and submit experimental reports.

4. Problem - solving Ability

Experimental teaching aims to cultivate students' problem - solving ability.

5. Three Experiments

Include K - Nearest Neighbors (KNN), PCA - SVM, and Convolutional Neural Network (CNN) experiments.

Weighting Details

Weights of different components

Class Participation Weighting

Class participation accounts for 10 points in the total score.

Homework Weighting

Homework is worth 20 points in the overall assessment.

Experiments Weighting

Experiments contribute 30 points to the final score.

Final Exam (or Capstone Project) Weighting

The final exam or capstone project is worth 40 points.

Book Recommendations

Suggested reading materials

Machine Learning (Tom M. Mitchell)

Published by China Machine Press in 2003, a classic in the field.

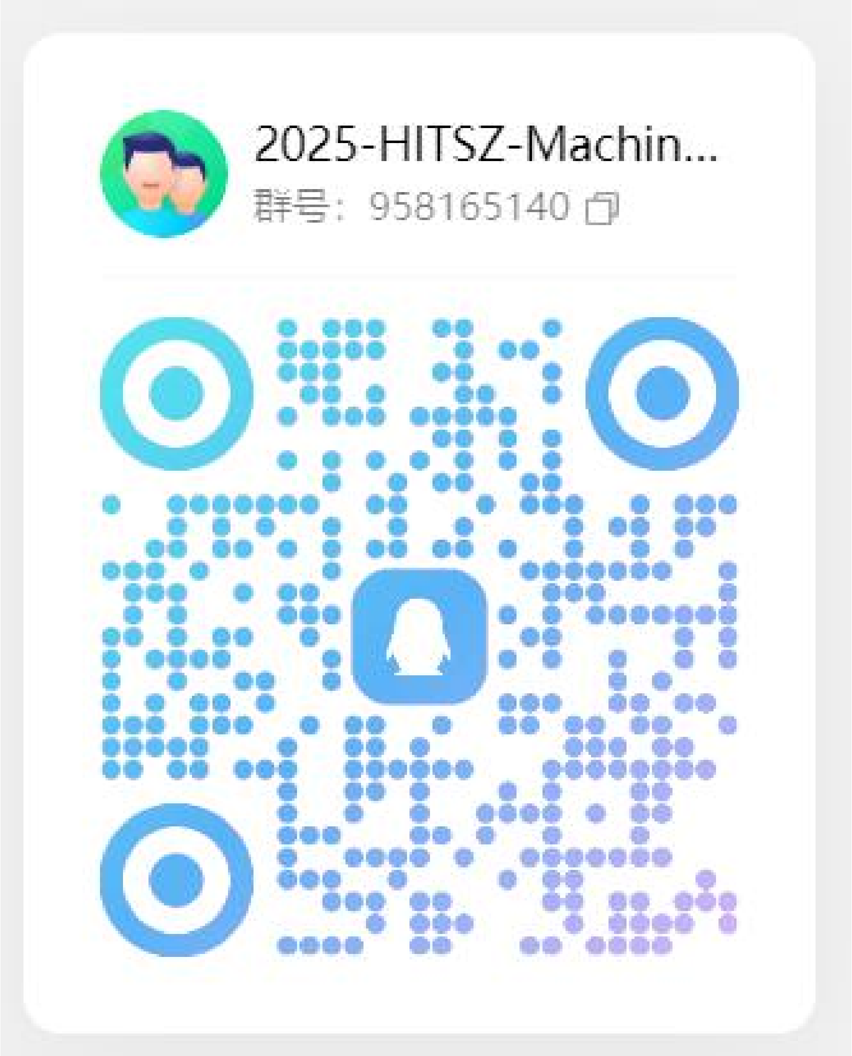
Pattern Recognition and Machine Learning (C. Bishop)

From Hans Publishers in 2006, covering pattern recognition aspects.

Deep Learning (Ian Goodfellow, Yoshua Bengio, Aaron Courville)

Released by MIT Press in 2016, focusing on deep learning techniques.

QQ Group ID



Introduction to Machine Learning

Understanding the Core Principles

What is Machine Learning?

Definition

Machine learning enables computers to learn from data and improve task performance.

Goal

It aims to build models for accurate predictions or decisions from input data.

Applications

Widely used in recommendation systems, image recognition, and medical diagnosis.

Common Algorithms

Includes linear regression, decision trees, SVMs, and neural networks.

Key Concepts

Grasping Fundamental Ideas

1

2

Model Accuracy

Ensures the model's predictions closely match real-world outcomes.

Overfitting and Underfitting

Overfitting occurs when a model is too complex, underfitting when it's too simple.

4

3

Training and Testing Data

Training data helps build the model, while testing data evaluates its accuracy.

Feature Selection

Involves choosing relevant input variables to enhance model performance.

Types of ML Algorithms

Exploring Different Approaches

Supervised Learning

Uses labeled data to learn mappings between inputs and outputs for prediction.

Unsupervised Learning

Analyzes unlabeled data to discover hidden patterns, structures, or relationships.

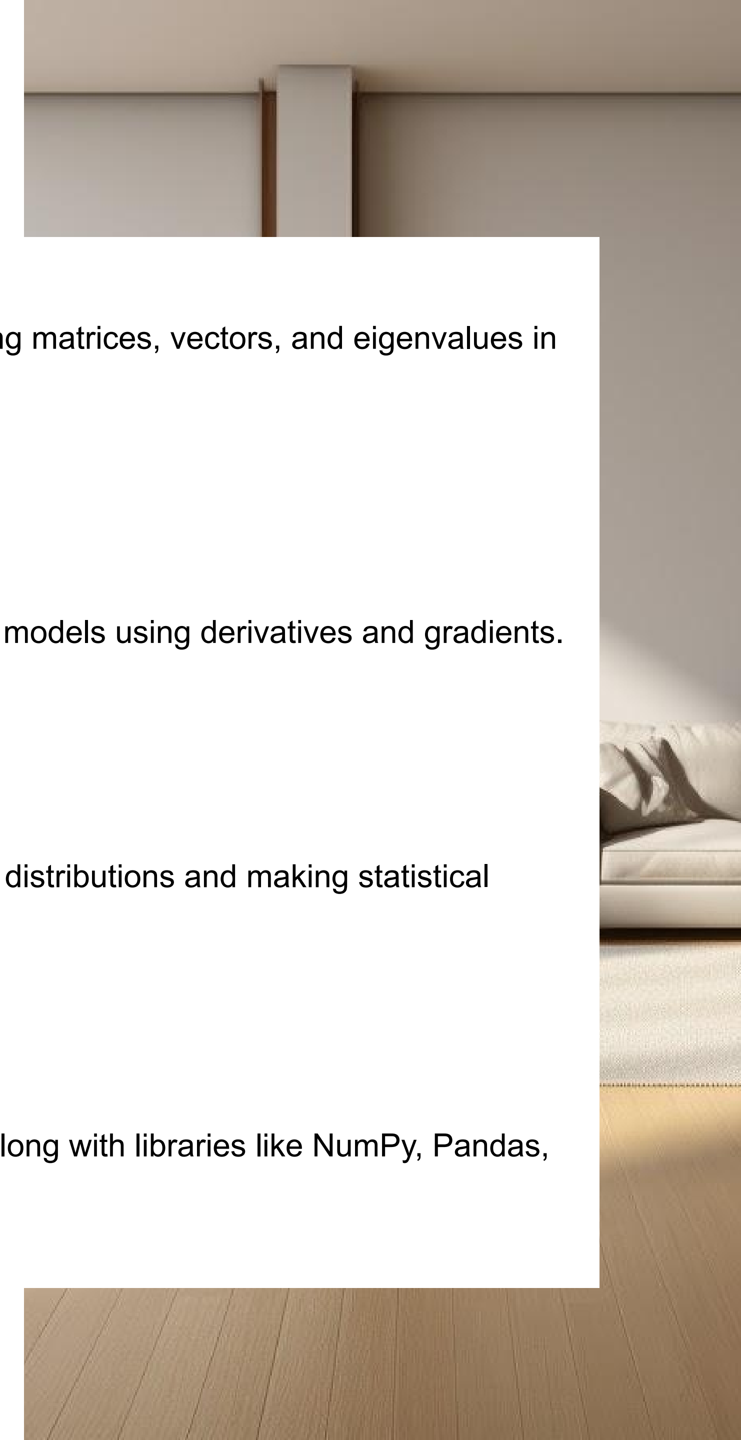
Reinforcement Learning

Learns through trial and error by receiving rewards or punishments for actions.

Semi-supervised Learning

Combines labeled and unlabeled data for more efficient model training.

Foundations: Math & Programming



Building the Necessary Skills

Linear Algebra

Crucial for understanding matrices, vectors, and eigenvalues in ML.

Calculus

Essential for optimizing models using derivatives and gradients.

Probability and Statistics

Helps in analyzing data distributions and making statistical inferences.

Programming Skills

Proficiency in Python, along with libraries like NumPy, Pandas, and scikit-learn.

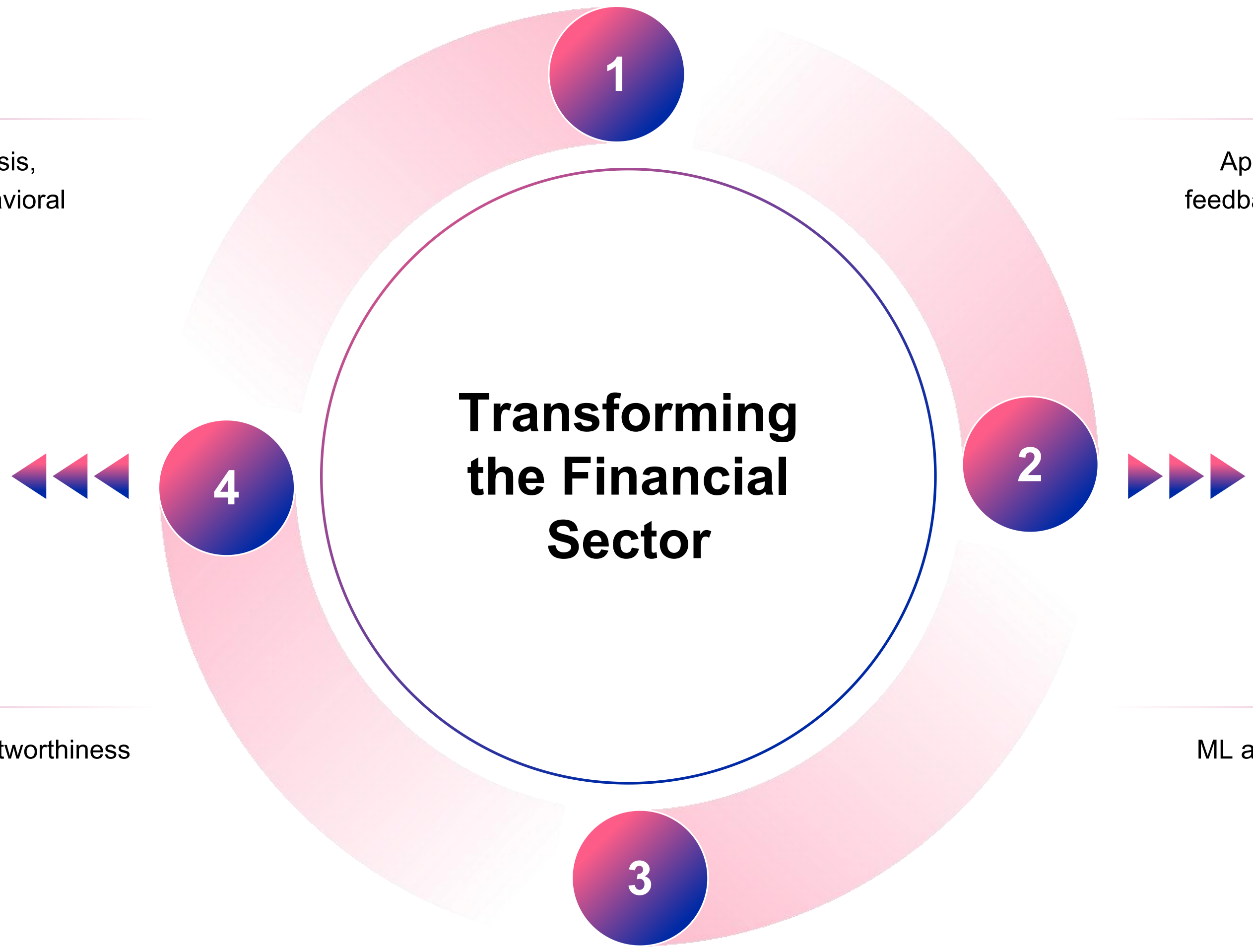
ML in Banking & Finance

Case 1: Careem (2024)

Implemented real-time transaction analysis, reducing fraud risk by 37% through behavioral anomaly detection.

Case 2: Monzo (2023)

Applied text data topic modeling to customer feedback, identifying savings patterns with 89% accuracy.



Credit Scoring

Analyzes various factors to assess creditworthiness and determine loan approvals.

Fraud Detection

ML algorithms can identify unusual transaction patterns to prevent financial fraud.

ML in Healthcare & Retail

Healthcare: Google DeepMind

Algorithm identifies eye diseases like diabetic retinopathy with high accuracy.

Retail: Wayfair

Uses ML for demand prediction and personalized product recommendations.

Enhancing Health and Shopping Experiences

Medical Image Analysis

Helps in diagnosing diseases by analyzing medical images such as X-rays and MRIs.

Inventory Management

Optimizes stock levels to reduce waste and improve customer satisfaction.

Major Industry Players in China

BAT giant ecology

Baidu (PaddlePaddle PaddlePaddle framework), Alibaba (Dharma Institute visual algorithm), Tencent (Youtu Lab) build a full stack technology solution, covering cloud computing to terminal applications.

Unicorn enterprise innovation

SenseTime (computer vision), Yitu Technology (medical AI), and Yuncong Technology (financial OCR) have formed technological barriers in vertical fields, with valuations exceeding billions of dollars.

Foreign R&D center landing

Microsoft Research Asia (MSRA) continues to output NLP breakthrough achievements, NVIDIA China has set up CUDA technology center to promote the optimization of localized tool chain.

AI-Powered Healthcare in China

Medical imaging diagnosis

Deep learning algorithms have achieved an accuracy rate of over 95% in CT/MRI image recognition. Shanghai Lianying Medical's AI assisted system can automatically label lung nodules and generate structured reports.

Personalized treatment plan

AI models based on patient genomic data and clinical records (such as Tencent Miying) can recommend targeted drug combinations, increasing the effectiveness of cancer treatment by 30%.

Remote monitoring system

wearable devices combined with AI prediction (such as Ping An Technology's ECG analysis) achieve real-time warning of home heart abnormalities, reducing emergency response time to within 5 minutes.

Drug development acceleration

Jingtai Technology adopts generative AI to design molecular structures, shortening the screening cycle of new drug compounds from traditional 4 years to 12 months, and reducing research and development costs by 60%.

Cross-Industry Collaboration in China

1

Smart Energy Management

Alibaba Cloud ET Industrial Brain collaborates with the power grid to predict regional electricity loads through ML, optimize wind/photovoltaic energy storage scheduling, and reduce the annual power abandonment rate by 15%.

2

Agricultural intelligence

Jifei Technology's unmanned aerial vehicles are equipped with computer vision, achieving a recognition accuracy of 98% for cotton field pests and diseases, reducing pesticide use by 40% while increasing production by 20%.

3

Financial risk control innovation

Ant Chain's federated learning technology breaks down cross institutional data silos, increasing the AUC value of small and micro enterprise credit evaluation models to 0.89 and reducing bad debt rates by 3.2 percentage points.

Enterprise Solutions

Business-Oriented ML Services

Indianic: Healthcare and Finance

Provides predictive analytics solutions for healthcare and finance sectors.

Leewayhertz: Generative AI

Offers generative AI solutions for various industries.

IBM Watson: Cognitive Computing

Delivers cognitive computing capabilities for businesses.

Amazon AI Services: Cloud-Based Solutions

Provides cloud-based AI services for scalable ML applications.

Alibaba AI Applications

01

Intelligent customer service

Alibaba Xiaomi is an AI customer service system developed by Alibaba, which can process massive user inquiries through natural language processing technology, achieve 24-hour online service, and greatly improve customer satisfaction and operational efficiency.

02

Supply chain optimization

Alibaba uses machine learning algorithms to analyze historical sales data and market trends, optimize inventory management and logistics delivery, reduce inventory backlog and improve delivery speed, supporting its massive e-commerce business.

03

Financial risk control

Ant Financial has built an intelligent risk control system based on machine learning models. By analyzing user behavior, transaction records, and other data, it can identify fraudulent transactions and credit risks in real time, ensuring the safety of the platform and users' funds.

ByteDance Recommendation Systems

Personalized content distribution

based on user behavior data (such as length of stay, likes, and comments), Tiktok and Toutiao's recommendation system, through collaborative filtering and deep learning models, realizes the content push of thousands of people and faces, and significantly improves user stickiness.

Real time feedback optimization

ByteDance recommendation algorithm can analyze user interaction data in real time and dynamically adjust recommendation strategies, such as quickly identifying user interest changes in short video platforms to reduce content fatigue.

Multimodal Content Understanding

Its AI system integrates multimodal data such as text, images, videos, and audio, and uses cross modal learning techniques to accurately understand content features, ensuring a high degree of matching between recommended content and user preferences.

Scikit-learn Tools

Popular Tools for ML Beginners

ROC Curves

Evaluate the performance of classification models by plotting true positive rate against false positive rate.

Decision Boundaries

Visualize the separation between different classes in a classification problem.

Partial Dependence Plots

Show the relationship between a feature and the model's prediction, ignoring other features.

Feature Importance Plots

Identify the most important features in a model for better understanding and interpretation.

GitHub Examples

Real-World Code Examples

Linear Regression Fitting Animation

Animates the process of fitting a linear regression model to data.

XOR Problem Visualization

Visualizes the solution to the XOR problem using neural networks.

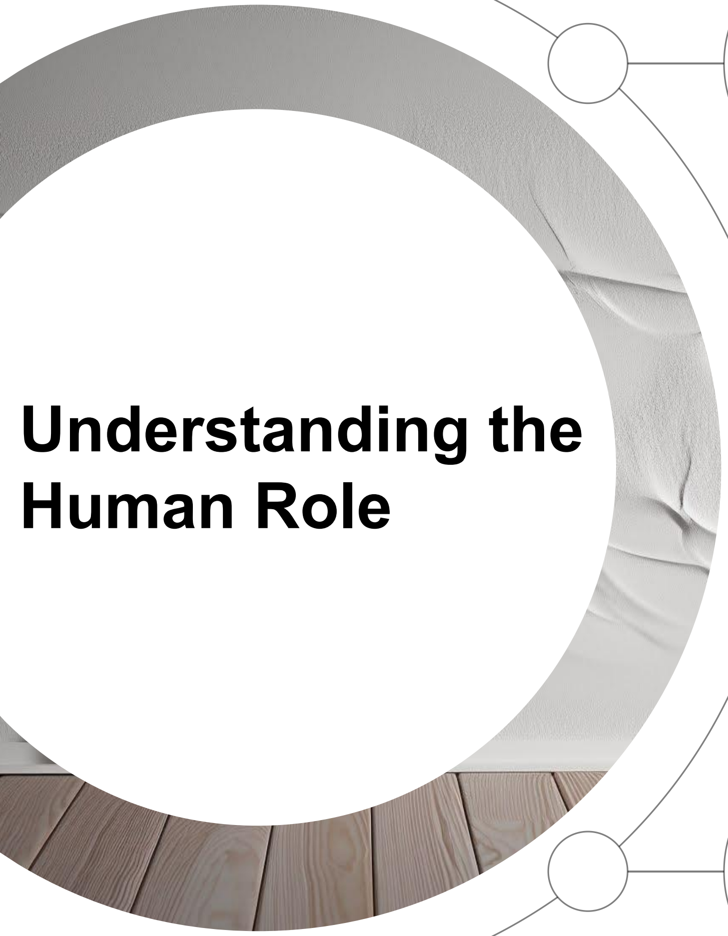
Clustering Algorithm Demonstration

Demonstrates the use of clustering algorithms on sample data.

Image Classification with CNN

Shows how to implement a convolutional neural network for image classification.

Misconception 1: "ML is Magic"



Understanding the Human Role

Human Intervention

Machine learning requires substantial human involvement in design, training, evaluation, and maintenance.

Ethical Considerations

Ensuring ethical implementation of ML models is crucial to prevent biases and misuse.

Model Interpretability

It's important to understand how models make decisions for better control and trust.

Data Quality

High-quality data is essential for accurate and reliable ML results.

Misconception 2: "ML is Easy"

Recognizing the Challenges

Technical Expertise

Developing robust ML solutions requires strong programming and mathematical skills.

Domain Knowledge

In-depth knowledge of the application domain is necessary for effective problem formulation.

Model Selection

Choosing the right algorithm and model architecture can be a complex task.

Data Preprocessing

Preparing data for ML models involves cleaning, normalization, and feature engineering.

Misconception 3: "Skip the Basics"

Emphasizing Fundamental Algorithms

Core Building Blocks

Classical algorithms like linear regression and decision trees are the foundation of modern ML.

Interpretability

Simple models often provide better interpretability than complex ones.

Efficiency

Basic algorithms can be more efficient in certain scenarios.

Problem Solving

Understanding fundamentals helps in solving complex ML problems effectively.

Recommended Courses

Top Online Learning Options

Udemy: "Machine Learning A-Z"

A comprehensive course covering various ML algorithms and techniques (43h, 4.5/5 rating).

Udemy: "Python Data Science and ML Hands-On"

Focuses on practical applications of ML using Python (33h, 4.5/5 rating).

Coursera: "Machine Learning by Andrew Ng"

A classic course taught by a renowned expert in the field.

edX: "Introduction to Machine Learning with Python"

Offers a hands-on approach to learning ML with Python.

GitHub Projects

Hands-On Practice Opportunities

"2025 ML and Data Science Starter"

A GitHub project providing hands-on practice with scikit-learn.

"ML Project Ideas"

Offers a list of project ideas for beginners to gain practical experience.

"Data Visualization with Matplotlib"

Teaches data visualization techniques using the Matplotlib library.

"Natural Language Processing Projects"

Includes projects on NLP using popular libraries like NLTK and spaCy.

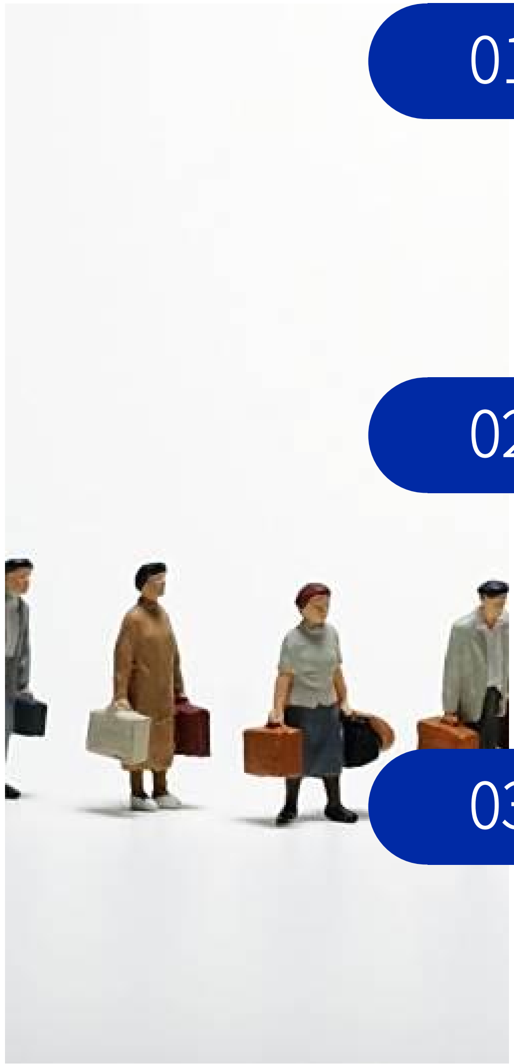
MOOC Platforms in China

Wide coverage of localized content: Chinese MOOC platforms such as China University MOOC and Xuetang Online provide a large number of Chinese machine learning courses, covering basic theories to industry applications, adapting to the domestic education system, and reducing language learning barriers.

Practical resources for school enterprise cooperation: The platform collaborates with companies such as Huawei and Alibaba Cloud to develop practical projects, providing real datasets and industry cases to help learners master skills that meet the needs of the domestic market.

National Excellent Course Certification: Among the "National First Class Courses" selected by the Ministry of Education, machine learning related courses account for a significant proportion, ensuring the authority and cutting-edge nature of teaching content.

International Platforms (Localized in China)



01

Coursera/edX Chinese interface

provide Chinese subtitles for machine learning and deep learning special courses, support Alipay payment, and some courses are equipped with Chinese learning community and teaching assistant services.

02

Udacity Nanodegree Localization

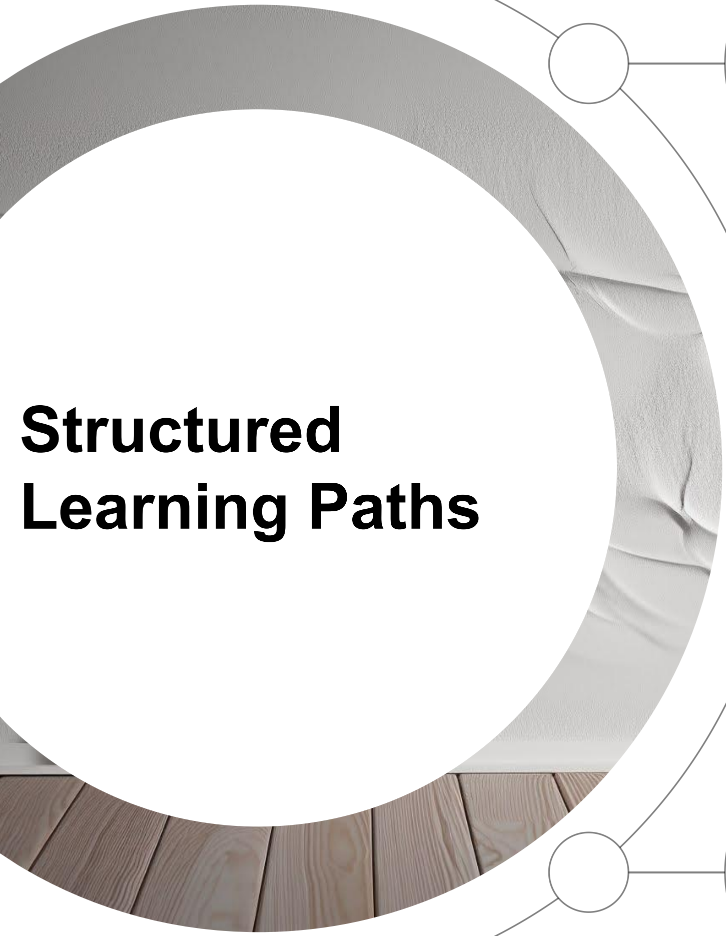
Collaborating with companies such as Tencent to launch Chinese language projects such as "Autonomous Driving" and "AI Engineers", incorporating Chinese traffic regulations and localized datasets into course cases.

03

Kaggle Competition Chinese Guide

The platform officially translates competition documents, establishes Chinese discussion forums, and reduces communication costs for domestic developers participating in international data science competitions.

Kaggle Roadmaps



Structured Learning Paths

2025 Kaggle Roadmap

Outlines a step-by-step learning path for beginners, including math, programming, and core algorithms.

Math Foundations

Covers essential mathematical concepts required for ML, such as linear algebra, calculus, and statistics.

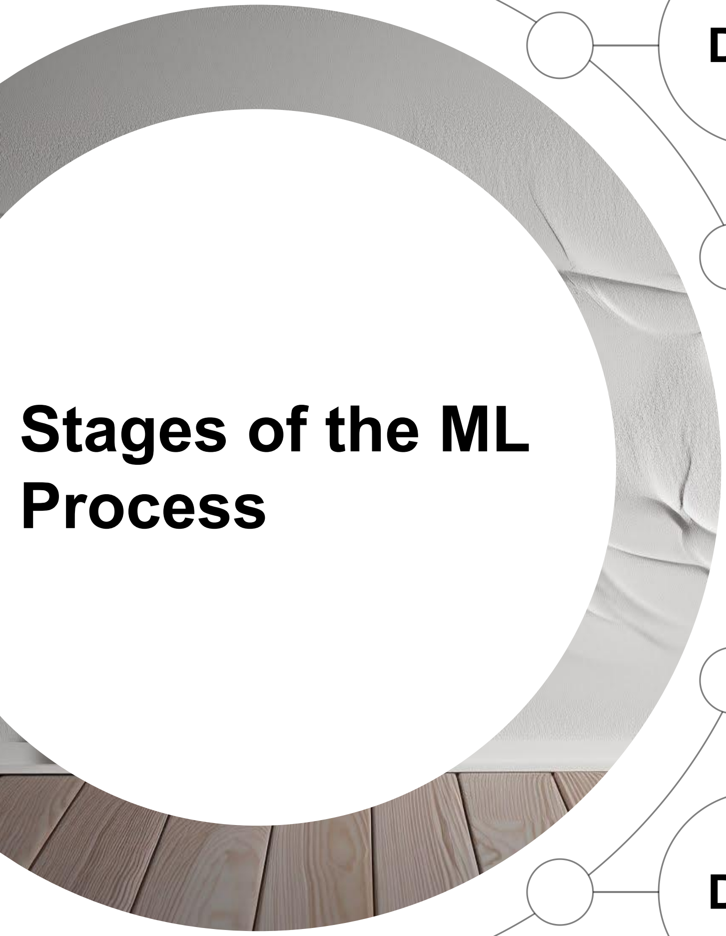
Programming Skills

Focuses on programming languages like Python and libraries like scikit-learn and TensorFlow.

Core Algorithms

Introduces basic ML algorithms and their applications.

Workflow Overview



Stages of the ML Process

Define the Goal

Clearly define the problem you want to solve with ML.

Data Collection

Gather relevant data for your ML project.

Data Preprocessing

Clean and prepare the data for model training.

Model Development

Select and train an appropriate ML model.

Deployment and Monitoring

Deploy the model and monitor its performance in real-world scenarios.

Data Collection & Exploration

Gathering and Understanding Data

S

Data Sources

Identify reliable sources of data for your project.

O

Data Quality Assessment

Evaluate the quality of data to ensure accurate results.

W

Data Exploration Techniques

Use techniques like descriptive statistics and visualization to explore data.

T

Data Cleaning

Remove noise, outliers, and missing values from the data.

Data Preprocessing & Feature Engineering

1. **Data Transformation**: Transform data into suitable formats for modeling.
2. **Feature Selection**: Select the most relevant features for the model.
3. **Feature Scaling**: Scale features to ensure equal importance in the model.
4. **Encoding Categorical Variables**: Convert categorical variables into numerical form for analysis.

Model Development & Validation

Building and Evaluating Models

Model Selection

Choose an appropriate ML model based on the problem and data.

Training the Model

Train the model using the preprocessed data.

Model Evaluation

Evaluate the performance of the model using metrics like accuracy and precision.

Hyperparameter Tuning

Optimize the model's performance by adjusting hyperparameters.

Deployment & Monitoring (MLOps)

Bringing Models to Production

Model Deployment	**20XX**	Deploy the trained model in a real-world environment.
Monitoring	**20XX**	Continuously monitor the model's performance and detect any issues.
Model Updates	**20XX**	Update the model as needed based on new data or changing requirements.
Feedback Loop	**20XX**	Implement a feedback loop to improve the model over time.

What is Deep Learning?

Defining Deep Learning

01

Definition

Deep learning is a subset of machine learning that uses neural networks with multiple layers.

02

Key Components

Includes artificial neurons, layers, and activation functions.

03

Learning Process

Learns hierarchical representations of data through multiple processing layers.

04

Applications

Used in image recognition, natural language processing, and speech recognition.

Neural Networks

Building Blocks of Deep Learning

01

Structure

Composed of input, hidden, and output layers connected by weights.

02

Activation Functions

Introduce non-linearity to the model, enabling it to learn complex patterns.

03

Training Process

Involves forward propagation and backward propagation with gradient descent.

04

Types

Includes feedforward, recurrent, and convolutional neural networks.

Training Deep Models

Techniques for Model Training

Optimizers	**Loss Functions**
Batch Size	**Regularization**

Algorithms like stochastic gradient descent used to update weights.

Measure the difference between predicted and actual outputs.

Prevents overfitting by adding constraints to the model.

Determines the number of samples used in each iteration of training.

Popular Deep Learning Frameworks

Tools for Deep Learning Development

TensorFlow

An open-source framework widely used for building and training deep learning models.

PyTorch

A popular framework known for its dynamic computational graph and ease of use.

Keras

A high-level neural networks API that runs on top of TensorFlow and other backends.

Caffe

Designed for deep learning applications in computer vision.

Pytorch Example

Healthcare

Advancing Medical Treatments

Drug Discovery

ML predicts drug interactions and accelerates the drug discovery process.

Personalized Medicine

Tailors treatments based on individual patient characteristics.

Telemedicine

Improves remote healthcare services using AI-powered tools.

Medical Device Innovation

Develops smart medical devices with AI capabilities.

Generative AI

Creating New Content

Text Generation

AI generates human-like text for various applications.

Image Synthesis

Creates realistic images from scratch or based on input.

Music Composition

Composes music in different styles using AI algorithms.

Video Generation

Generates videos with high-quality content and motion.

Industrial

Revolutionizing Manufacturing

01.	**Predictive Maintenance**	Predicts equipment failures to reduce downtime.
02.	**Quality Control**	Ensures product quality using real-time monitoring.
03.	**Supply Chain Optimization**	Optimizes supply chain operations for efficiency.
04.	**Smart Factories**	Automates manufacturing processes with AI.

Data Preparation for Machine Learning

Advanced Techniques, Python Implementation & Best Practices

Data Preparation : The Foundation of ML Success

- Data preparation accounts for 60-80% of ML project time (LinkedIn, 2025)
- **Impact:** Poor data quality leads to 50%+ model performance degradation
- **Core objectives:** Accuracy, completeness, consistency, relevance
- **Scope:** Collection, cleaning, feature engineering, scaling, validation

Core Data Preparation Steps

01 Data Collection & Validation

Define requirements, verify accuracy/completeness (LinkedIn, 2025).

02 Data Cleaning

Remove invalid values, handle missing data, filter outliers (Oracle, 2022).

03 Feature Engineering

Select/transform features to enhance model performance.

04 Feature Scaling/Normalization

Standardize ranges for consistent training (Google Developers, 2025).

Feature Engineering & Trends

Dimensionality Reduction

PCA/t-SNE reduce noise while retaining key features (Devinterview, 2024).

Feature Selection

Correlation matrices, Lasso regression, Elastic Net for high-dimensional data.

Polynomial/Interaction Features

Capture non-linear relationships (e.g., age*income).

Trend: AI-Driven Automation

Tools like ThoughtSpot (2025) auto-format and filter data, significantly reducing manual effort and accelerating the feature engineering pipeline.

Strategies for Handling Missing Values

1. Avoid Deletion: Drop rows/columns only for extreme missingness and low-importance features (Kaggle, 2024).
2. Imputation Methods: Mean (normal data), median (skewed data), mode (categorical data); KNN/MICE for complex datasets.
3. Time-Series Specific: Forward/backward fill or interpolation for sequence continuity.
4. Special Cases: 0 for 'no sales' or binary flags for event occurrence.

Data Cleaning Best Practices

02 Define Quality Goals

Identify issues (duplicates, missing values) and set targets (e.g., 'reduce missing values by 20%').

04 Comprehensive Profiling

Use pandas/missingno to visualize patterns (random vs. systematic missingness).

06 Automated Validation

Tools like ThoughtSpot's AI assist flag inconsistencies during collection.

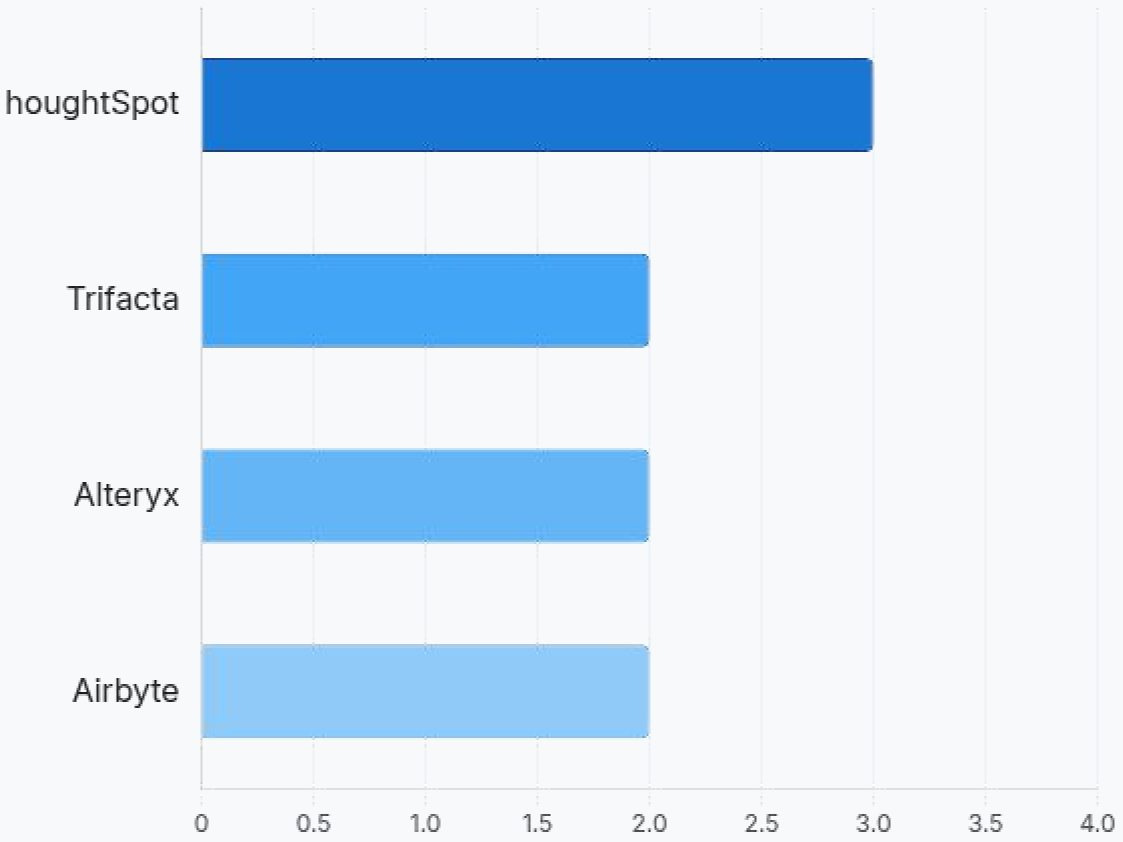
08 Iterative Refinement

Continuously update pipelines to adapt to new data patterns (LinkedIn, 2025).

Automated Data Preparation Tools

Tool	Key Features	Best For
ThoughtSpot	AI assist, SQL/Python/R integration	End-to-end prep/analysis
Trifacta	Visual transformation, handles messy data	Unstructured multi-source data
Alteryx	Drag-and-drop, cloud connectors	No-code/low-code workflows
Airbyte	550+ connectors, vector DB integration	Large-scale ML pipelines

Tool Feature Coverage



Source: The 6 best data preparation tools for analysts in 2025 (ThoughtSpot, 2025)

Real-World Case Studies

01 Software Defect Prediction

Used Smallset Timelines (R package) to track edits and validate preprocessing decisions, improving defect identification accuracy.

02 Wealth Management KPI Analysis

Automated R scripts cleaned access logs, filtered outliers, and generated dashboards for marketing optimization (GitHub, 2024).

Data Cleaning with Python: Pandas Implementation

Core operations: Remove duplicates, handle missing values, standardize formats.

```
# Remove duplicates
df.drop_duplicates(inplace=True)

# Standardize phone numbers
df['phone_number'] = df['phone_number'].str.replace(r'[^\\w]', '', regex=True)

# Split address into components
df[['street', 'state', 'zip']] = df['address'].str.split(',', expand=True)
```

Best practice: Use pandas profiling (missingno) for pattern visualization (Kaggle, 2024)

Missing Value Handling: KNN Imputation

- Limitations of basic methods (mean/median): Distorts feature correlations
- KNN imputation theory : Use k-nearest neighbors' values to preserve relationships

```
from sklearn.impute import KNNImputer

imputer = KNNImputer(n_neighbors=2)
data_imputed = imputer.fit_transform(data)
```

Advantages: Superior to simple imputation for complex datasets (GeeksforGeeks, 2025)

Outlier Detection Techniques

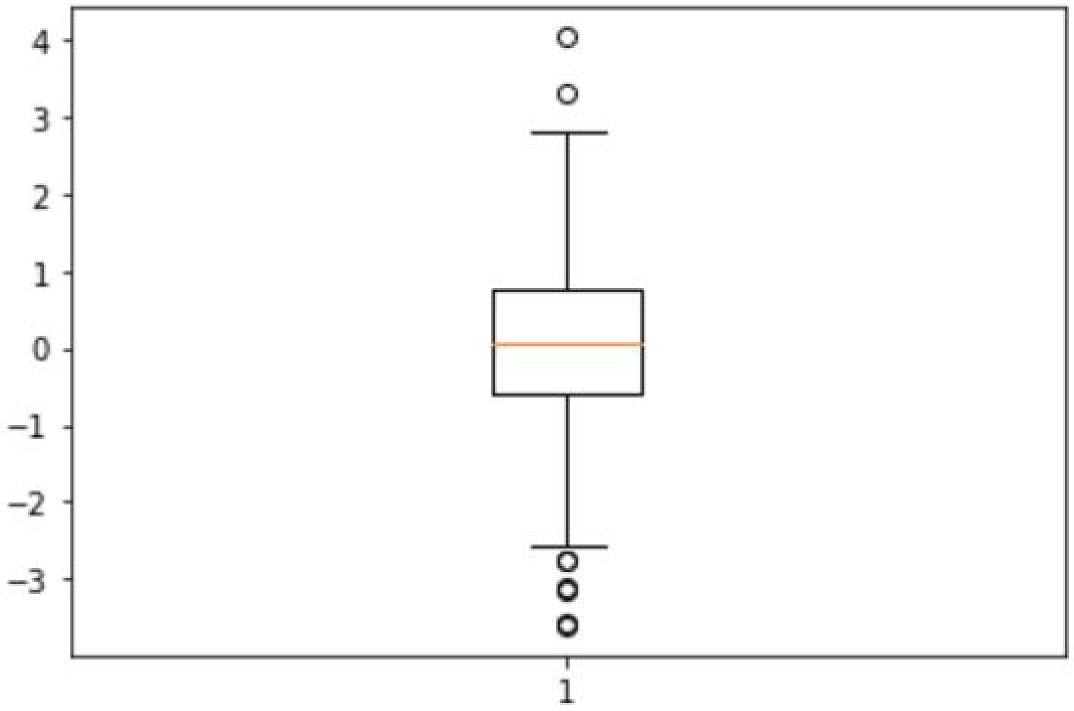
Statistical methods: Z-score ($|z| > 3$), Dixon's Q-test

ML methods: One-Class SVM, Isolation Forest

Python example (Z-score):

```
z_scores = [(x - mean)/std for x in data]
outliers = [x for x in z_scores if abs(x) > 3]
```

A box plot defines the range of normal data via its whiskers (typically based on 1.5 times the interquartile range, or IQR); discrete points outside the whiskers' range are displayed as outliers.



Feature Engineering: PCA & Scaling

PCA for dimensionality reduction:

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)
```

Preserves 95%+ variance while reducing noise.

Feature scaling:

- StandardScaler: Mean=0, std=1 (for SVM/KNN)
- RobustScaler: Resistant to outliers (using IQR)

Best Practices & Automated Tools

AI-driven automation: **40% reduction in manual effort** (ThoughtSpot, 2025)

Tool comparison:

- **ThoughtSpot:** AI assist, SQL/Python integration
- **Trifacta:** Visual transformation for messy data
- **Alteryx:** No-code workflows with cloud connectors
- **Airbyte:** 550+ connectors for large-scale pipelines

MLOps integration: DVC for versioning, MLflow for reproducibility

Ethical Considerations in Data Preparation

Bias mitigation: Inclusive data collection, diverse validation teams

Privacy protection: GDPR/CCPA compliance, anonymization techniques

Fairness: Detect feature bias using Aequitas tool (Ethical AI, 2024)

Transparency: Document preprocessing steps for stakeholder trust

Risk: Unrepresentative datasets lead to discriminatory model outcomes

Conclusion & Future Trends for Data Preparation

- **Key takeaways:** Rigorous preprocessing = robust ML models; Python tools + automation = efficiency
- **Future trends:** AutoML preprocessing, real-time data validation, federated learning for privacy
- **Acknowledgements:** Scikit-learn documentation, ThoughtSpot 2025 reports, Kaggle community

References

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[2] GeeksforGeeks. (2025). *Python Imputation using KNNImputer*. https://www.geeksforgeeks.org/python-imputation-using-the-knnimputer/

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Model Selection for Machine Learning

Model Selection Foundations: Bias-Variance Tradeoff

- **Bias**: Error from oversimplified assumptions (e.g., linear models for nonlinear data) → underfitting.

- **Variance**: Error from sensitivity to training data fluctuations → overfitting.

- **Tradeoff**: Total Error = $\text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$.

- **Key Insight**: Optimal complexity aligns with data characteristics to balance generalization and fit.

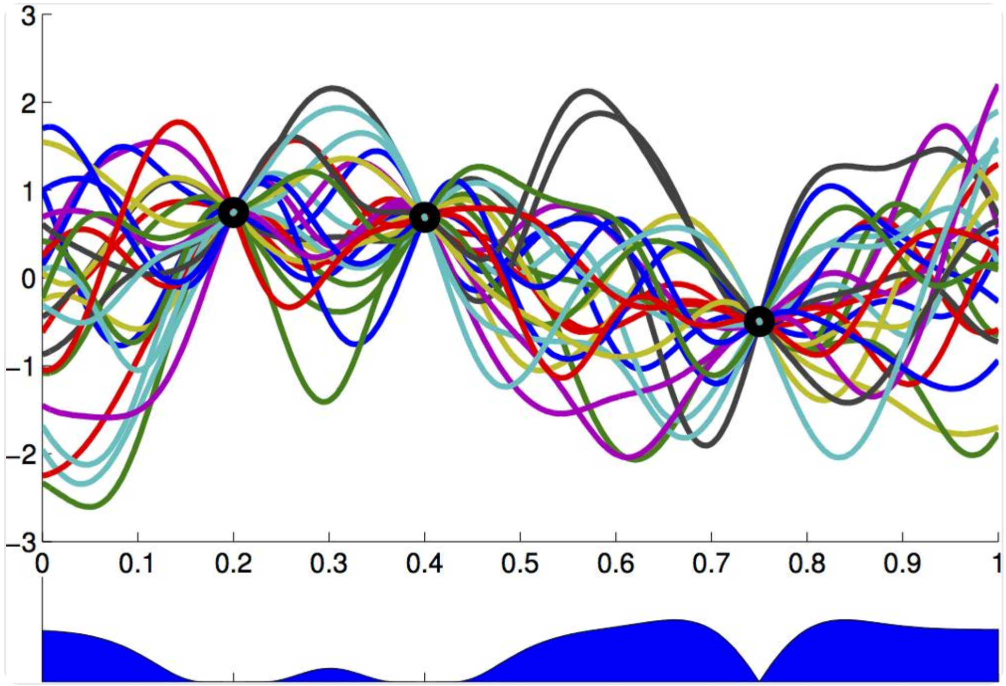
Model Selection Techniques

- **Traditional**: Grid search, randomized optimization.

  Advanced: Bayesian optimization (adaptive tuning)

- **Trend**: Bayesian nonparametrics (dynamic complexity)

- **Workflow**: Simple → complex → validate.



Evaluation Metrics Comparison

- **Classification:** Accuracy, Precision, Recall ; F1-score (imbalanced data).
- **Regression:** MAE, MSE/RMSE, R^2 (variance explained)
- **LLM-Specific (2025):** Perplexity (text coherence), Functional Pass Rate (code)

Cross-Validation Methods

Purpose: Assess generalization by evaluating on multiple subsets, reducing overfitting risk

Common Types:

- **k-Fold CV:** Split into k folds; train on k-1, validate on 1.
- **Stratified k-Fold:** Preserves class distribution (critical for imbalanced data)
- **Time Series CV:** Consecutive temporal splits.

Implementation: Scikit-learn tools: `cross_val_score()`, `StratifiedKFold()`

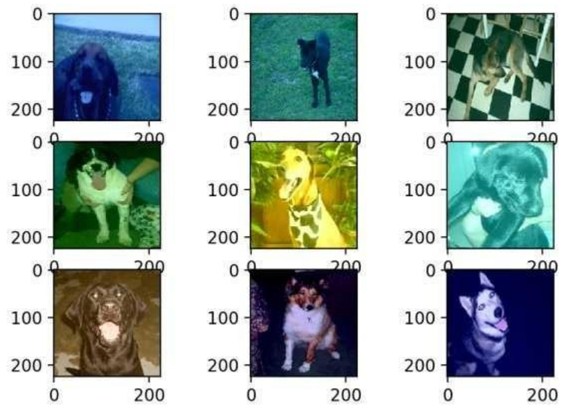
Overfitting Prevention Strategies

- **Hyperparameter Tuning:** Adjust learning rate, batch size, etc.

- **Data Augmentation:** Generate synthetic samples

- **Resampling:** SMOTE, RandomUnderSampler

- **Regularization:** L1/L2 penalties, dropout, early stopping.



Real-World Case Studies

Flood Risk Assessment (Charlotte, NC, 2025):

Models: Logistic regression, XGBoost, Random Forest.

Result: Logistic regression outperformed (5.55% very high risk, 52.20% very low risk)

Chronic Kidney Disease Prediction (2024):

Models: LR, RF, DT, NB, XGBoost.

Result: XGBoost best (ROC-AUC 0.9689, accuracy 93.29%)

Emerging Trends

- **AutoML Tools:** Cerebros (NAS), Nano AutoML Cloud
- **Imbalanced Data:** Hellinger distance, cost-sensitive loss
- **Bayesian Nonparametrics:** Flexible models adapting complexity
- **LLM Selection:** Query-aware models (SelectLLM)
- **Sustainability:** Carbon intensity-aware DNN inference[

Conclusion for Model Selection

- **Summary:** Model selection requires balancing bias-variance tradeoff, robust validation, and alignment with use cases.
- **Future Directions:** Sustainability metrics integration, Bayesian nonparametrics advances, LLM-specific frameworks.
- **Acknowledgements:** Supported by scikit-learn, AutoML community, and 2024-2025 case studies.

Training and Evaluation for Machine Learning

1. Introduction to ML Training & Evaluation

- Core Definitions: Training = Model learns patterns from data; Evaluation = Assesses performance and reliability.
- Importance: Prevents overfitting and ensures generalization to unseen data.
- Trends: Widespread AutoML adoption, multi-dimensional evaluation systems, and large-scale model training.
- Key Process: Data Preparation → Algorithm Selection → Training → Validation → Deployment.

2. Supervised Learning: Methods & Trends

Enhanced Model Architectures:

Transformers: Expanded to image classification/time series (long-range dependency handling).

Ensemble Methods: Stacking, boosting, voting, bagging (e.g., Random Forest) for robustness.

GBM: Optimized for multi-type data processing, reducing overfitting.

Key Advances:

Transfer Learning: Fine-tuning pre-trained Transformers on small datasets (data-scarce domains).

Active Learning: Selective data querying to reduce annotation needs.

3. Unsupervised Learning: Innovations & Applications

Core Applications:

- Customer segmentation: Identifying hidden groups from purchasing behavior.
- Raw data analysis: Exploring unstructured data (e.g., customer emails) for insights.

Case Studies:

- DCAEC Model: Deep Convolutional Autoencoder for leukocyte clustering (Balanced Accuracy 91.9%).
- Turbulence Super-Resolution: CycleGAN reconstructs high-res flow fields from filtered data (performar near supervised models).
- Interpretability: Grad-CAM reveals cluster-specific visual patterns.

4. Reinforcement Learning Applications

Human Feedback Alignment:

- Align-SLM: Reinforcement Learning from AI Feedback (RLAIF) improves semantic consistency of text-free speech models.

Robotics & Real-World Environments:

- DR-MARL: Distributed Robust Multi-Agent RL optimizes Amazon warehouse chute mapping, reducing package cycles.

Long Horizon LLM Agents:

- Loop Algorithm: Memory-efficient RL training for interactive LLM agents (32B parameter model outperforms OpenAI O1 by 15%).

5. Evaluation Metrics: Classification Tasks

Precision: $\text{Precision} = \frac{TP}{TP+FP}$ (Proportion of correct positive predictions)

Recall: $\text{Recall} = \frac{TP}{TP+FN}$ (Ability to identify all actual positives)

F1 Score: $F1 = \frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$ (Balance between precision and recall)

Accuracy: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$ (Overall correctness)

Application: Interpreting the Confusion Matrix in medical diagnosis, where the risk of False Negatives is critical.

6. Evaluation Metrics: Regression Tasks

MSE: MSE = $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ (Penalizes large errors)

RMSE: RMSE = $\sqrt{\text{MSE}}$ (Interpretable in original units)

MAE: MAE = $\frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|$ (Average absolute error)

R^2: Proportion of variance explained (0-1, higher is better)

Example: For sales predictions with actual [200,300,350,400] and predicted [207,290,360,390], MSE = 87.25

7. Evaluation Metrics: Clustering Tasks

Internal Metrics:

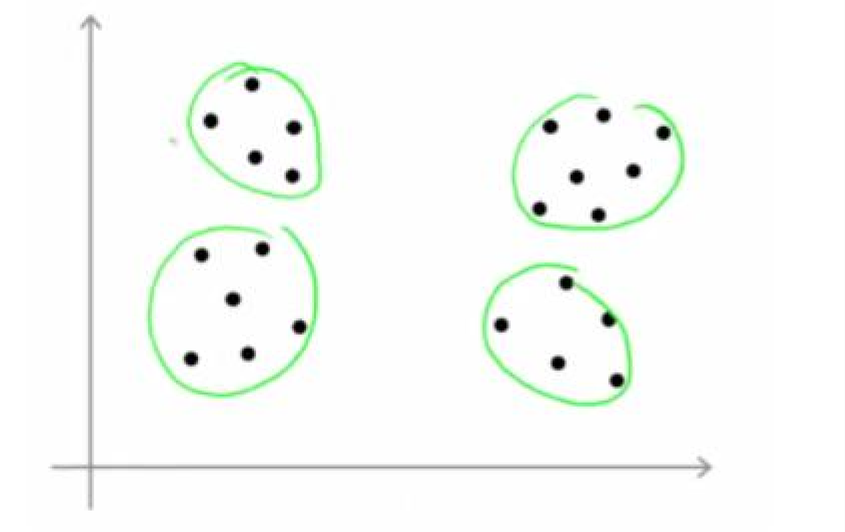
WCSS (Elbow Method): $\text{WCSS} = \sum_{i=1}^{n} \sum_{j=1}^{k} |x_i - c_j|^2$ (Minimizes within-cluster variance)

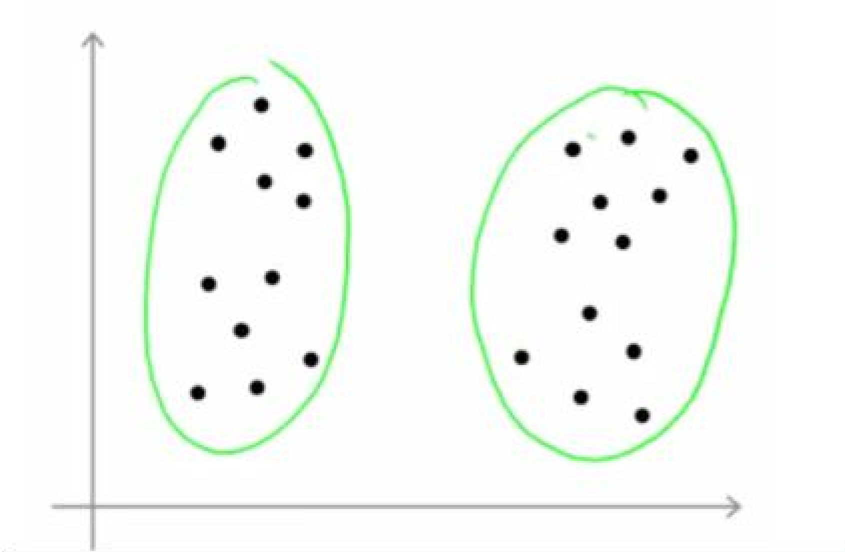
Silhouette Score: $s = \frac{b-a}{\max(a,b)}$ (Measures cluster separation, -1~1)

Calinski-Harabasz Index: Between-cluster variance / Within-cluster variance (Higher is better)

Challenges & Advances:

Curse of dimensionality and embedding space variability. 2024 frameworks reduce metric misuse via external validation alignment.





8. Case Study 1: MNIST & CIFAR-10

MNIST Dataset:

- Handwritten digits (0-9), 60k training / 10k test images.
- 3-layer ANN performance: 98-99% accuracy, training loss up to 430.

CIFAR-10 Dataset:

- 10 classes (airplane, car, etc.), 50k training / 10k test.
- 5-layer ANN performance: 95-98% accuracy, training loss up to 600.

Optimization Techniques:

CNN, Transfer Learning (VGG-16), HOG+SVM improve performance. Accuracy is positively correlated with computational cost.

9. Case Study 2: Drug Discovery

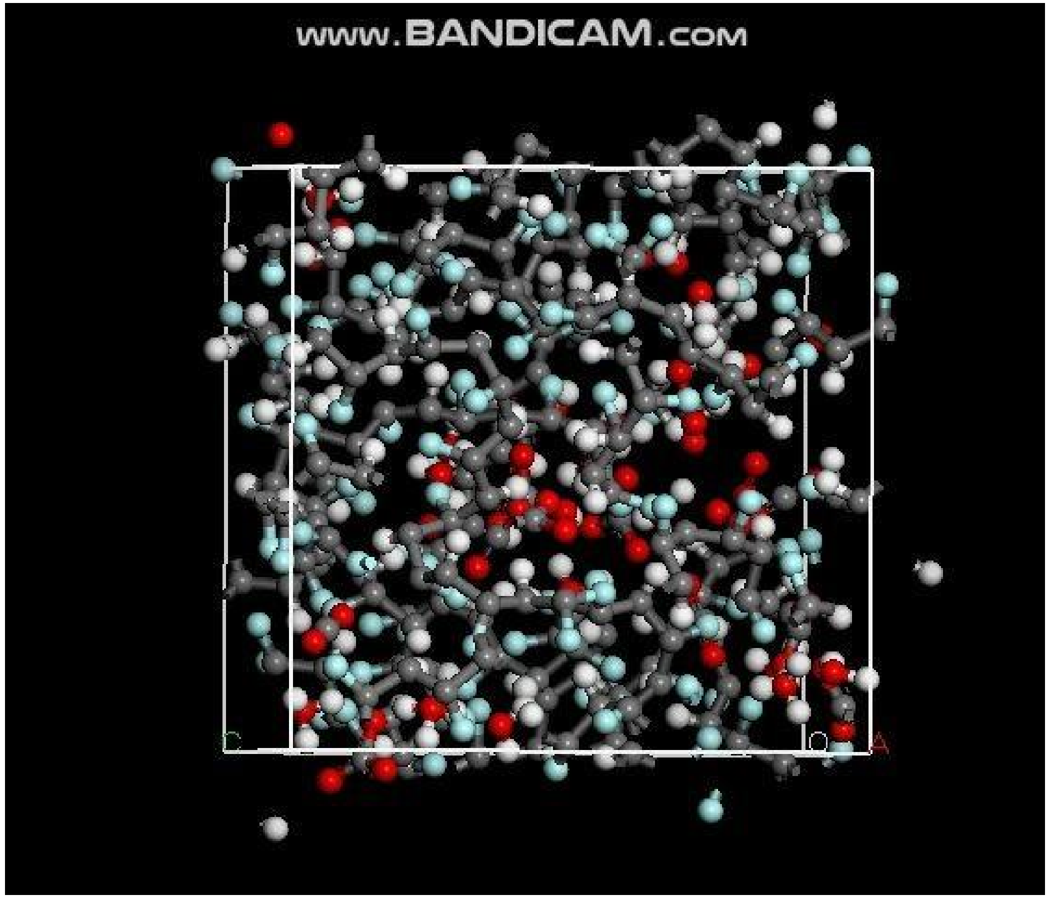
Goal: Identify ALK inhibitors from a natural product library.

ML Pipeline:

- Ensemble tree models, kernel methods, neural networks, and structural dynamics.
- Input: Molecular structures, binding affinity data; Output: 6 potential inhibitors.

Validation Results:

- MD simulations (>100 ns) show stable binding at Glu105, Met107, Asp178 residues.
- Minimal conformational fluctuations confirm binding efficacy.



10. Case Study 3: Energy Project Management

- Goal: Predict Variation Orders (VOs) in energy construction projects.
- Methodology:
  - Dataset: Historical project data (activities, timelines, costs).
  - Models: 6 regression algorithms (Random Forest, Gradient Boosting, etc.).
- Key Findings:
  - Significant non-linear impact of project activities (linear models are insufficient).
  - Gradient Boosting is optimal: $MAE=12.3, R^2 = 0.84$, supporting proactive risk mitigation.

11. Conclusion & Future Trends for Traning and Evaluation

Core Conclusions:

- Training requires adaptive algorithms (Transformers, AutoML) for efficiency.
- Evaluation needs multi-dimensional metrics (task-specific + generalization).
- Case studies validate ML's value in healthcare, energy, and computer vision.

Future Directions:

- AutoML Deployment: Deployment time from weeks to hours.
- Multi-dimensional evaluation systems: Beyond accuracy (e.g., LLM teaching quality).
- Generalization enhancement: Synthetic datasets for stress-testing unseen scenarios.
- Quantum-enhanced RL: Optimizing complex multi-agent systems.

Machine Learning Metrics: Fundamentals, Applications, and Advanced Trends

Classification vs. Regression Tasks

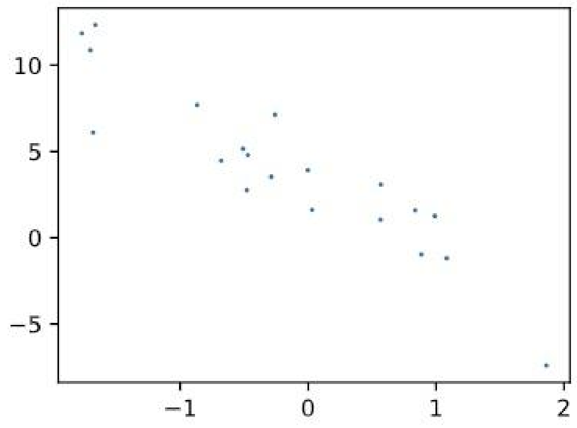
Classification: Predict discrete class labels (e.g., spam detection).

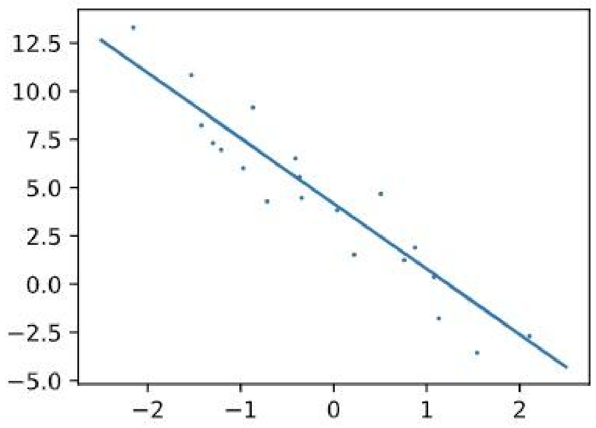
- Key metrics: Accuracy, Precision, Recall, F1, ROC-AUC.

Regression: Predict continuous numerical values (e.g., house price prediction).

- Key metrics: MSE, RMSE, MAE, R^2.

Selection Criteria: Depends on target variable type and business objectives.





Confusion Matrix: Foundation of Classification Metrics

Definition: 2x2 matrix summarizing model predictions vs. true labels.

Core Components:

True Positive (TP): Correctly predicted positive cases.

True Negative (TN): Correctly predicted negative cases.

False Positive (FP): Incorrectly predicted positive (Type I error).

False Negative (FN): Incorrectly predicted negative (Type II error).

Example: Fire Severity Analysis)

	Predicted Positive	Predicted Negative
Actual Positive	TP = 70	FN = 10
Actual Negative	FP = 5	TN = 312

F1 Score: Balancing Precision and Recall

Definition: Harmonic mean of precision and recall, balancing both metrics.

Formula:

$$\frac{2 \times Precision \times Recall}{Precision + Recall}$$

Key Advantage: Mitigates imbalance between precision and recall.

Ideal Use Cases:

Imbalanced datasets (e.g., disease screening) .

When both false positives and false negatives are costly.

Interpretation: Range [0,1], higher values indicate better balance.

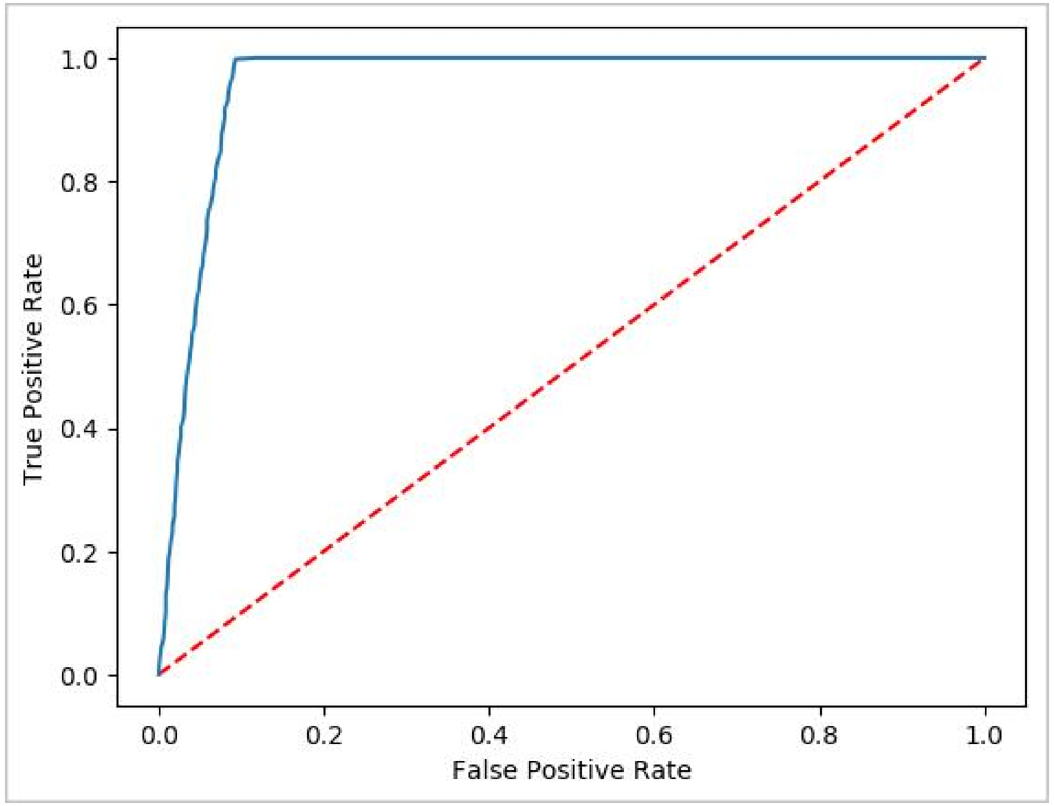
ROC Curve and AUC: Model Discrimination Ability (Part 1)

ROC Curve: Plots the True Positive Rate (TPR, also known as Recall) versus the False Positive Rate (FPR) across various classification thresholds.

AUC (Area Under Curve): Quantifies the overall performance of a model.

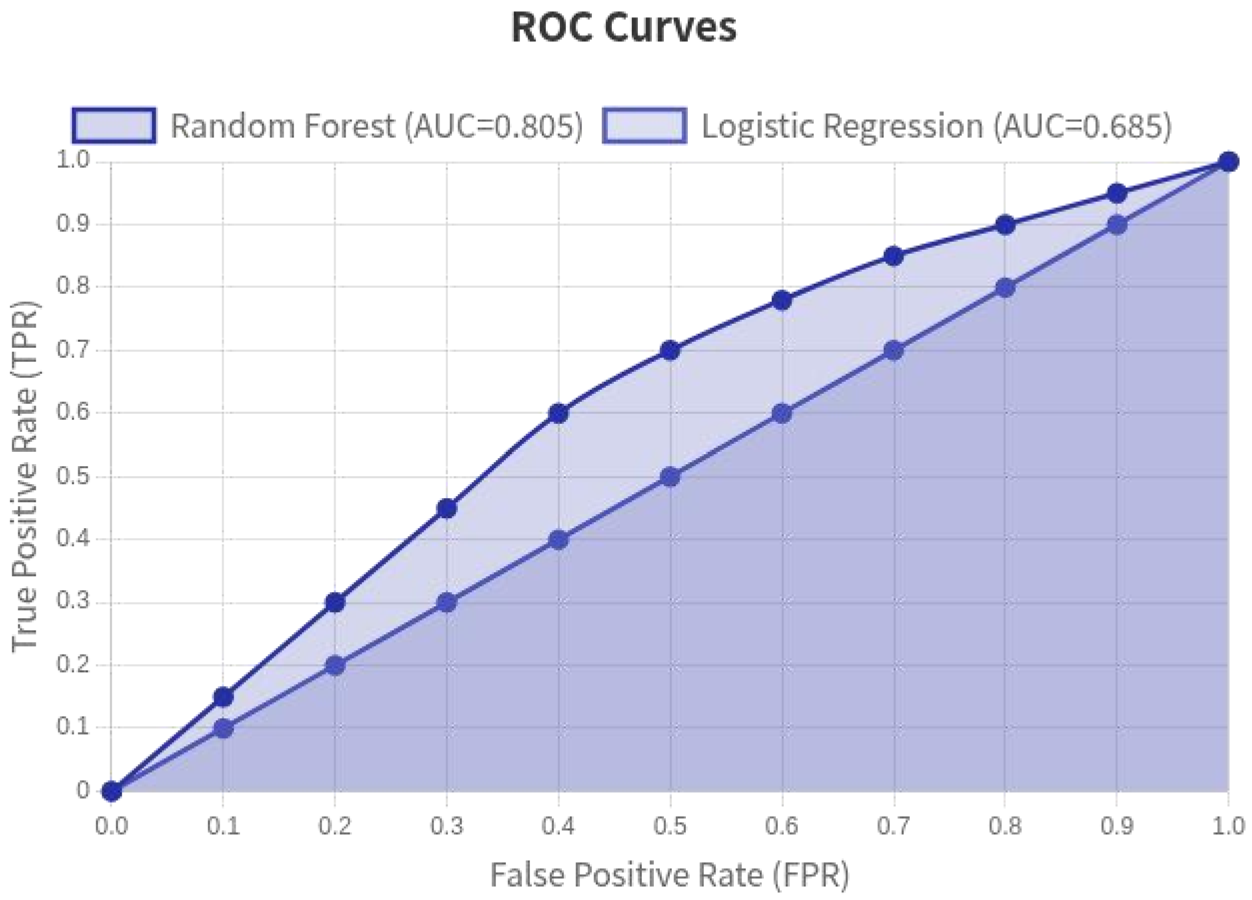
AUC = 0.5: Indicates a random guess (no discrimination).

AUC > 0.5: Indicates performance better than random guessing; the closer the AUC is to 1, the stronger the model's discrimination.



ROC Curve and AUC: Model Discrimination Ability (Part 2)

- **Example:** Penicillin efficacy model (AUC>0.5 indicates predictive ability) ([13]).
- **Limitation:** Less reliable for imbalanced datasets (use PR-AUC instead) ([45615]).
- Implementation: Use `sklearn.metrics.roc_auc_score` for computation ([scikit-learn docs]).



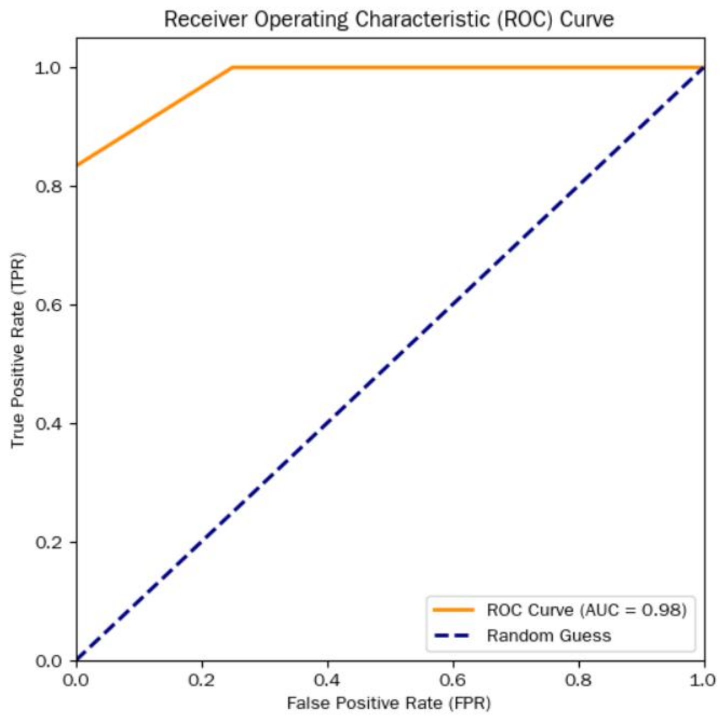
```python
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt
import numpy as np

# Example: True labels (0 = negative, 1 = positive) and predicted probabilities
y_true = np.array([0, 0, 1, 1, 1, 0, 1, 0, 1, 1])  # True classes
y_score = np.array([0.1, 0.2, 0.3, 0.4, 0.7, 0.25, 0.8, 0.3, 0.9, 0.6])  # Predicted scores/probabilities

# Calculate ROC curve and AUC
fpr, tpr, thresholds = roc_curve(y_true, y_score)
roc_auc = auc(fpr, tpr)

# Plot the ROC curve
plt.figure(figsize=(6, 6))
lw = 2  # Line width
plt.plot(
    fpr, tpr,
    color='darkorange',
    lw=lw,
    label=f'ROC Curve (AUC = {roc_auc:.2f})'
)
# Add the "random guessing" reference line
plt.plot([0, 1], [0, 1], color='navy', lw=lw, linestyle='--', label='Random Guess')

# Format plot
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate (FPR)')
plt.ylabel('True Positive Rate (TPR)')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc="lower right")
plt.show()
```

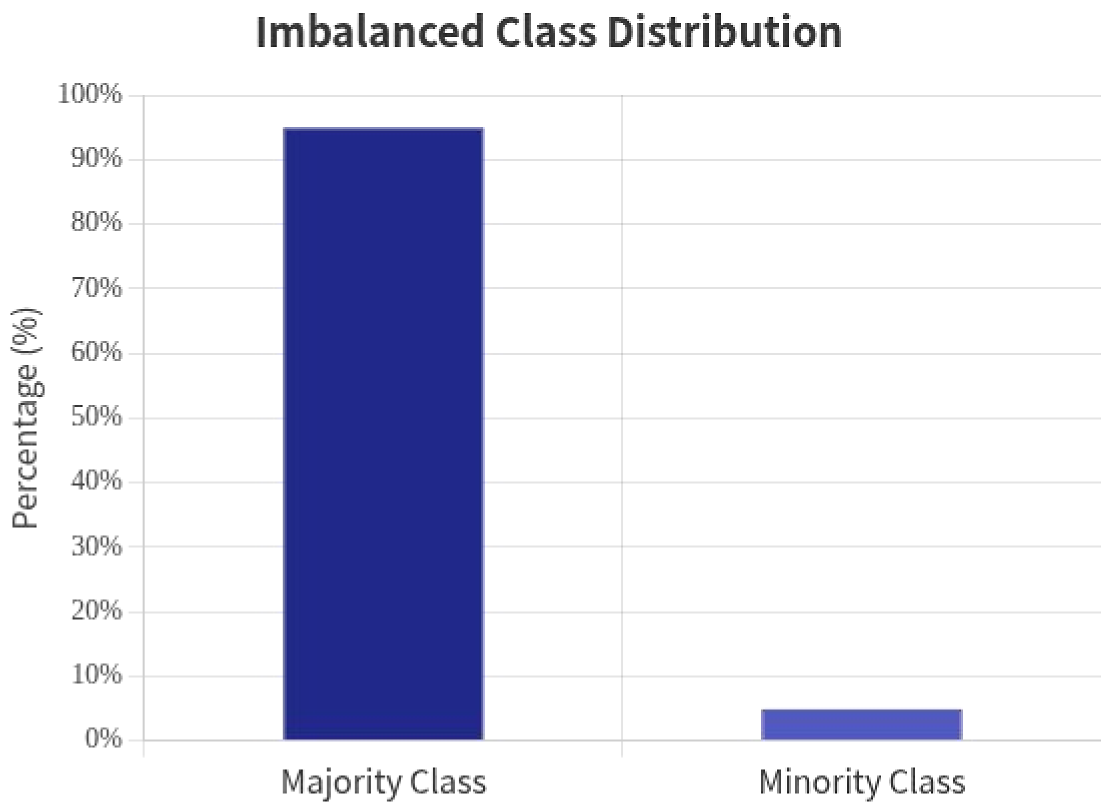


ROC Code Produces:

- A smooth curve showing TPR vs. FPR across all thresholds.
- A dashed line for "random guessing" (to contextualize performance).
- The AUC value (e.g., AUC = 0.98) to summarize how well the model distinguishes positive vs. negative classes.

Challenges in Imbalanced Datasets

- Problem: Majority class dominates (e.g., 99% 'good' products).
- Impact: Misleading high accuracy (e.g., 99% accuracy by predicting 'all good' but missing all defects).
- Root Causes: Sampling bias, rare events (e.g., fraud, rare diseases).
- Example: Snowfall prediction with 96% 'no snow' samples; accuracy fails to reflect predictive power .



Recommended Metrics for Imbalanced Data

- MCC (Matthews Correlation Coefficient): Balances TP/TN/FP/FN; ranges [-1,1], 1=perfect.

.

- PR-AUC (Precision-Recall AUC): Focuses on positive class performance; better than ROC-AUC for imbalance.
- F1 Score: Harmonic mean of precision/recall (balances false positives/negatives) .
- Avoid: Accuracy, geometric mean (G-mean) in extreme imbalance .
- Industry Example: Kaggle credit card detection (PR-AUC recommended over ROC-AUC) .

Multi-Class Classification Metrics

- **Macro Averaging:** Calculate metric per class, then average (equal weight to all classes).
- **Micro Averaging:** Aggregate TP/FP/FN across classes, then compute metric (weights by class size).
- **Weighted Averaging:** Macro-style per-class metrics weighted by class frequency.
- Use Cases:

  - **Macro:** Balanced classes, need per-class performance.
  - **Micro:** Imbalanced classes, overall performance focus.
- **Example:** 3-class fraud detection (macro for rare fraud types, micro for overall rate) .

Regression Metrics: Error and Explained Variance (Part 1)

MSE (Mean Squared Error)

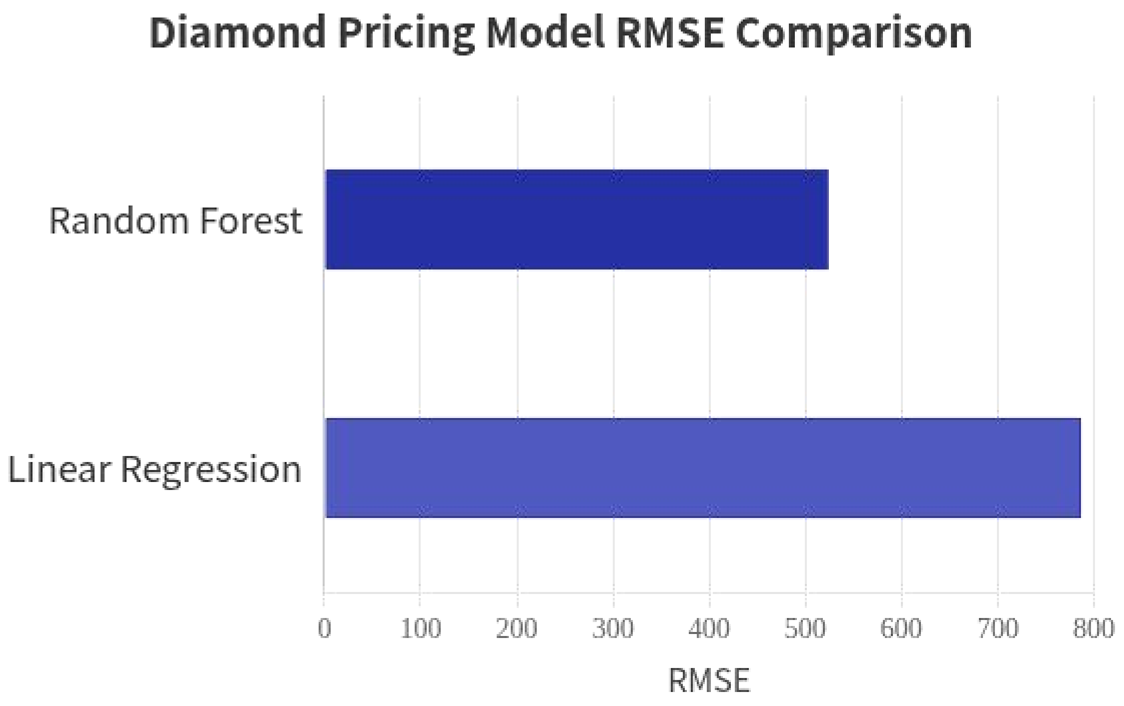
$$\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Use case: Penalize large errors (e.g., house prediction).

RMSE (Root MSE)

$$\sqrt{MSE}$$

Same unit as target. Example: Diamond pricing model (RF regression RMSE=523.50).



Regression Metrics: Error and Explained Variance (Part 2)

- MAE (Mean Absolute Error):
  - **Use Case:** Suitable for scenarios where outliers are present, such as in demand forecasting.

- R^2 (Coefficient of Determination):
  - **Example:** In a diamond pricing model, an R^2 value of 0.985 indicates that 98% of the variance in the data is explained by the model.

Industry Case Studies: Metrics in Practice

- ## Diamond Pricing Prediction (Regression):

  - **Model:** Random Forest regression.
  - **Metrics:** RMSE=523.50, R^2=0.985.

- ## Fraud Detection (Imbalanced Classification):

  - **Challenge:** 1% fraud rate.
  - **Metrics:** PR-AUC=0.85, F1=0.78.

- ## Ad Prediction System:

  - **Offline metric:** Log loss.
  - **Online metric:** Click-Through Rate (CTR) improvement by 15% .

Metrics Selection Framework: A Decision Guide

Classification Task:

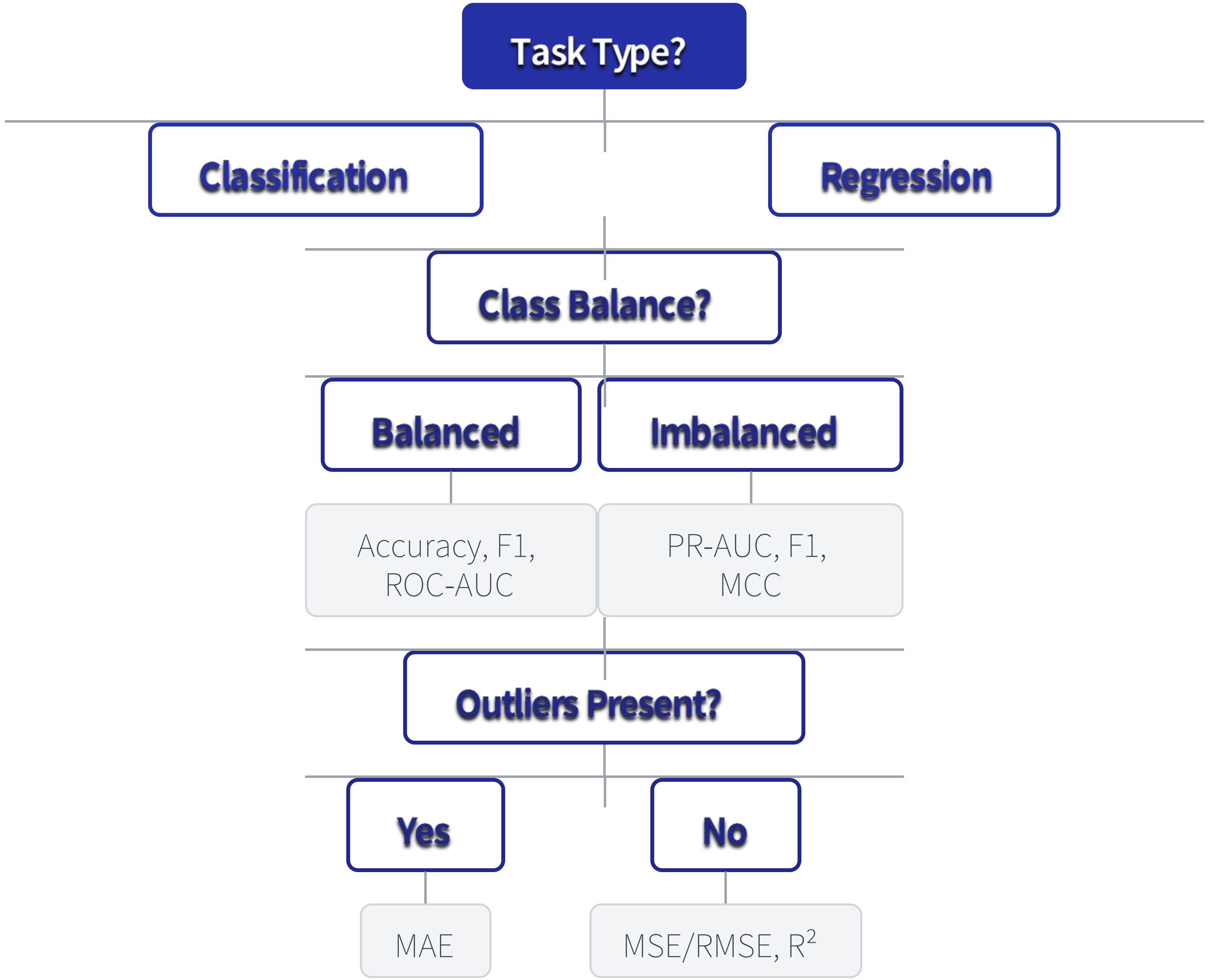
- Balanced classes: Accuracy, F1, ROC-AUC.
- Imbalanced classes: PR-AUC, F1, MCC.
- Multi-class: Macro/micro F1, weighted precision.

Regression Task:

- Outliers present: MAE.
- Penalize large errors: MSE/RMSE.
- Explain variance: R^2.

Key Question: What is the cost of false positives vs. false negatives?

ML Metrics Selection Decision Tree



Trends in Machine Learning Metrics

- Multi-Dimensional Evaluation: Beyond accuracy to include efficiency (latency), energy consumption, and ethical safety (bias detection).

- Industry Adoption: 64% of US enterprises using ML; SMB adoption growing 10% YoY (Medical 28%, Finance 21% leading).

- Dynamic Metrics: Real-time evaluation for online learning models with streaming data .

- Generative AI Metrics: New benchmarks for consistency, realism, and alignment in generative models.

Conclusion for Machine Learning Metrics

- Metrics must align with task type (classification/regression) and data characteristics (balance).
- Imbalanced datasets require specialized metrics (PR-AUC, F1, MCC) over accuracy.
- Multi-class scenarios demand careful averaging (macro/micro/weighted).
- **Trend focus:** Multi-dimensional evaluation, real-time metrics, and ethical considerations.
- **Best Practice:** Combine quantitative metrics with business impact analysis.

Computation Example for Machine Learning Metrics

A Step-by-Step Guide with Binary Classification Examples

Confusion Matrix

Definition: A table summarizing classification model performance by comparing predicted vs. actual labels for binary classification.

Structure: 2x2 matrix with True Positives (TP), True Negatives (TN), False Positives (FP), False Negatives (FN).

Sample Data: TP=45, TN=35, FP=10, FN=10; Total samples=100.

Confusion Matrix for Binary Classification

	Predicted Positive	Predicted Negative
Actual Positive	45 (TP)	10 (FN)
Actual Negative	10 (FP)	35 (TN)

Accuracy

1. Definition: Proportion of correctly classified samples out of total samples.
2. Formula: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$
3. Computation: $\frac{45+35}{100} = \frac{80}{100} = 0.80$ (80%)
4. Interpretation: Suitable for balanced datasets; misleading for imbalanced classes (e.g., rare diseases).

Precision

Definition: Proportion of predicted positive samples that are actually positive (Positive Predictive Value).

Formula: $Precision = \frac{TP}{TP+FP}$

Computation: $\frac{45}{45+10} = \frac{45}{55} \approx 0.818$ (81.8%)

Interpretation: Critical for minimizing false positives (e.g., spam detection, where false positives annoy users).

Recall

Definition: Proportion of actual positive samples correctly identified (Sensitivity/True Positive Rate).

Formula: $\text{Recall} = \frac{TP}{TP+FN}$

Computation: $\frac{45}{45+10} = \frac{45}{55} \approx 0.818$ (81.8%)

Interpretation: Essential for minimizing false negatives (e.g., disease diagnosis, where missing positives risks lives).

F1 Score

Definition: Harmonic mean of precision and recall, balancing both metrics.

Formula: $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$

Computation: $2 \times \frac{0.818 \times 0.818}{0.818 + 0.818} \approx 2 \times \frac{0.669}{1.636} \approx 0.818 \quad (81.8\%)$

Interpretation: Ideal when precision and recall are equally important (e.g., balanced binary classification tasks).

ROC Curve: Definition & Axes

Definition: Plots True Positive Rate (TPR) vs. False Positive Rate (FPR) across classification thresholds.

Axes:

- X-axis: $FPR = \frac{FP}{FP+TN}$ (False Positive Rate)
- Y-axis: $TPR = Recall = \frac{TP}{TP+FN}$ (True Positive Rate)

Threshold Impact: Lower thresholds → more positives → higher TPR and FPR.

Ideal Curve: Closer to top-left corner (high TPR, low FPR) = better performance.

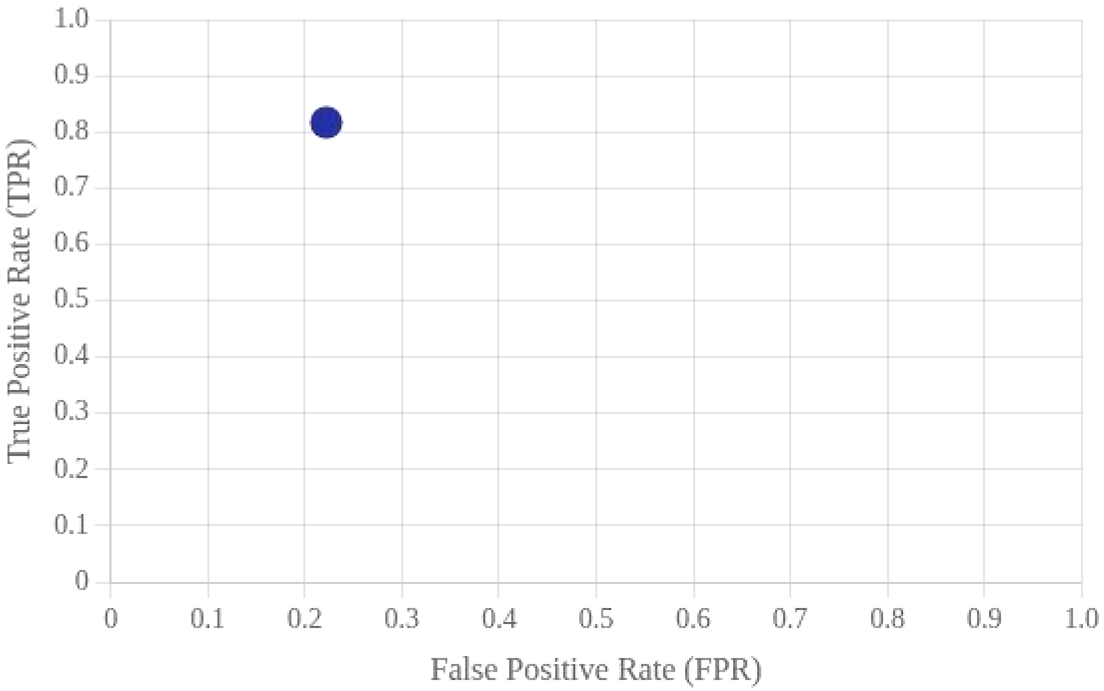
ROC Curve: Computation Example

Sample Data: TP=45, TN=35, FP=10, FN=10 (base threshold).

FPR Calculation: $\frac{FP}{FP+TN} = \frac{10}{10+35} = \frac{10}{45} \approx 0.222$ (22.2%)

TPR Calculation: $\frac{TP}{TP+FN} = \frac{45}{45+10} = \frac{45}{55} \approx 0.818$ (81.8%)

Plot Point: (0.222, 0.818); Additional threshold points form the complete curve.



AUC (Area Under ROC Curve)

- Definition: Quantifies the area under the ROC curve; measures model’ s ability to distinguish classes.
- Range: 0.5 (random guess) to 1.0 (perfect discrimination).
- Sample Data Context: With ROC point (0.222, 0.818) and typical threshold variation, AUC ≈ >0.8 (strong discrimination).
- Use Case: Model comparison; higher AUC = better overall performance across thresholds.

Summary of Machine Learning Metrics

Confusion Matrix: Foundation (TP=45, TN=35, FP=10, FN=10).

Accuracy: 80% (overall correctness; balanced datasets).

Precision: 81.8% (minimize false positives).

Recall: 81.8% (minimize false negatives).

F1 Score: 81.8% (balance precision-recall).

ROC/AUC: (0.222, 0.818) point; AUC>0.8 (strong class discrimination).

Key Takeaway: Choose metrics based on error cost (e.g., false negatives vs. false positives).